WORKED SOLUTION

STEP II 2005, Question 7

Pure • Vectors and Circular Motion

STEP II 2005, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q7

The key idea

Each position vector has the form $\cos(\text{angle})\,\mathbf{a} + \sin(\text{angle})\,\mathbf{b}$, and when $\mathbf{a}$ and $\mathbf{b}$ are perpendicular and of equal length this traces out a circle in the plane of $\mathbf{a}$ and $\mathbf{b}$. Once we know $|OP|$ and $|OQ|$ are constant, the angle POQ comes straight from the scalar product, and product-to-sum formulae tidy it into a single cosine. The last part is then an inequality for one cosine over two full periods.

Solution

Part (i)

Write $\mathbf{p}$ and $\mathbf{q}$ for the position vectors of P and Q.

For P, $\mathbf{p} = \cos t\,\mathbf{i} + \sin t\,\mathbf{j}$, so $|\mathbf{p}| = 1$ and $\mathbf{p}$ lies in the xy-plane. So P moves round the circle of radius 1, centre O, in the plane $z = 0$, anticlockwise when viewed from above (from the positive z-axis), with constant angular speed 1 (so speed 1). It starts at $(1, 0, 0)$ and makes exactly one revolution as t runs from 0 to 2π.

For Q, put $s = t + \frac{1}{4}\pi$ and

$$\mathbf{a} = \tfrac{3}{2}\mathbf{i} + \tfrac{3\sqrt{3}}{2}\mathbf{k}, \qquad \mathbf{b} = 3\mathbf{j}, \qquad \text{so that} \qquad \mathbf{q} = \cos s\,\mathbf{a} + \sin s\,\mathbf{b}.$$

Now $|\mathbf{a}|^2 = \frac{9}{4} + \frac{27}{4} = 9$, $|\mathbf{b}|^2 = 9$ and $\mathbf{a} \cdot \mathbf{b} = 0$. Hence

$$|\mathbf{q}|^2 = 9\cos^2 s + 9\sin^2 s + 2\cos s \sin s\,(\mathbf{a} \cdot \mathbf{b}) = 9,$$

and $\mathbf{q}$ always lies in the plane through O spanned by $\mathbf{a}$ and $\mathbf{j}$. That plane contains the y-axis and the direction $\mathbf{i} + \sqrt{3}\,\mathbf{k}$, so its equation is $z = \sqrt{3}\,x$; its normal $-\sqrt{3}\,\mathbf{i} + \mathbf{k}$ makes an angle $\arccos\frac{1}{2} = \frac{1}{3}\pi$ with $\mathbf{k}$.

So Q moves round a circle of radius 3, centre O, in the plane $z = \sqrt{3}\,x$, which contains the y-axis and is inclined at 60° to the plane of P's circle. It too has angular speed 1 (speed 3) and makes one revolution, rotating from the direction of $\mathbf{a}$ towards $\mathbf{j}$, and its phase is always $\frac{1}{4}\pi$ ahead of P's: at $t = 0$ it is at angle $\frac{1}{4}\pi$ from $\mathbf{a}$, at $\frac{3}{2\sqrt{2}}(1,\, 2,\, \sqrt{3})$.

Part (ii)

Since $|\mathbf{p}| = 1$ and $|\mathbf{q}| = 3$,

$$\cos\theta = \frac{\mathbf{p}\cdot\mathbf{q}}{3} = \frac{\frac{3}{2}\cos t\cos s + 3\sin t\sin s}{3} = \tfrac{1}{2}\cos t\cos s + \sin t\sin s,$$

where $s = t + \frac{1}{4}\pi$. The product-to-sum formulae, with $t + s = 2t + \frac{1}{4}\pi$ and $s - t = \frac{1}{4}\pi$, give

$$\cos t\cos s = \tfrac{1}{2}[\cos(2t + \tfrac{1}{4}\pi) + \cos\tfrac{1}{4}\pi],$$
$$\sin t\sin s = \tfrac{1}{2}[\cos\tfrac{1}{4}\pi - \cos(2t + \tfrac{1}{4}\pi)].$$

Writing $C = \cos(2t + \frac{1}{4}\pi)$ and using $\cos\frac{1}{4}\pi = \frac{\sqrt{2}}{2}$,

$$\begin{aligned}\cos\theta &= \tfrac{1}{4}(C + \tfrac{\sqrt{2}}{2}) + \tfrac{1}{2}(\tfrac{\sqrt{2}}{2} - C)\\ &= \tfrac{\sqrt{2}}{8} + \tfrac{\sqrt{2}}{4} - \tfrac{1}{4}C = \tfrac{3\sqrt{2}}{8} - \tfrac{1}{4}\cos(2t + \tfrac{1}{4}\pi),\end{aligned}$$

as required.

Part (iii)

? **Ask yourself** How do we turn $\theta \geq \frac{1}{4}\pi$ into a condition on $\cos\theta$?

Cosine is strictly decreasing on $[0, \pi]$, and both θ and $\frac{1}{4}\pi$ lie in that interval, so

$$\theta \geq \tfrac{1}{4}\pi \iff \cos\theta \leq \cos\tfrac{1}{4}\pi = \tfrac{\sqrt{2}}{2} = \tfrac{4\sqrt{2}}{8}.$$

By part (ii) this is

$$\tfrac{3\sqrt{2}}{8} - \tfrac{1}{4}\cos(2t + \tfrac{1}{4}\pi) \leq \tfrac{4\sqrt{2}}{8} \iff \cos(2t + \tfrac{1}{4}\pi) \geq -\tfrac{\sqrt{2}}{2}.$$

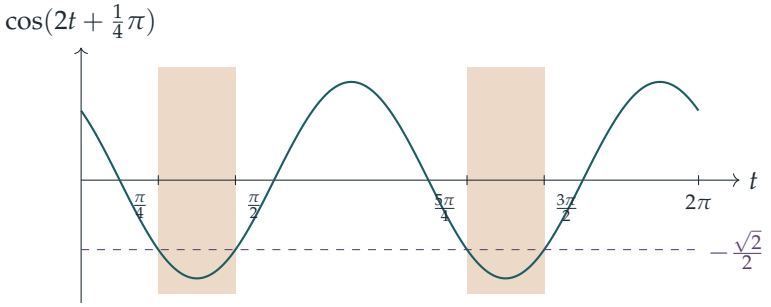
Let $u = 2t + \frac{1}{4}\pi$. As t runs over $[0, 2\pi]$, u runs over $[\frac{1}{4}\pi, \frac{17}{4}\pi]$, an interval of length 4π: exactly two periods of $\cos u$. In any period, $\cos u < -\frac{\sqrt{2}}{2}$ exactly on an open interval of length $\frac{1}{2}\pi$ (between $\frac{3}{4}\pi$ and $\frac{5}{4}\pi$, modulo 2π). Here the failing values are

$$u \in (\tfrac{3}{4}\pi, \tfrac{5}{4}\pi) \cup (\tfrac{11}{4}\pi, \tfrac{13}{4}\pi) \iff t \in (\tfrac{1}{4}\pi, \tfrac{1}{2}\pi) \cup (\tfrac{5}{4}\pi, \tfrac{3}{2}\pi),$$

each of length $\frac{1}{4}\pi$ in t. So $\theta < \frac{1}{4}\pi$ for a total time of $\frac{1}{2}\pi$, and the total time for which $\theta \geq \frac{1}{4}\pi$ is

$$2\pi - \tfrac{1}{2}\pi = \tfrac{3}{2}\pi,$$

as required.



The shaded strips are where the curve dips below $-\frac{\sqrt{2}}{2}$, that is, where $\theta < \frac{1}{4}\pi$.

Checks

- At $t = 0$: $\mathbf{p} = \mathbf{i}$ and $\mathbf{q} = \frac{3}{2\sqrt{2}}(\mathbf{i} + 2\mathbf{j} + \sqrt{3}\,\mathbf{k})$, so directly $\cos\theta = \frac{3}{2\sqrt{2}} \cdot \frac{1}{3} = \frac{\sqrt{2}}{4}$. The formula gives $\frac{3\sqrt{2}}{8} - \frac{1}{4} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4}$. They agree.
- $\cos\theta$ lies between $\frac{3\sqrt{2}}{8} - \frac{1}{4} \approx 0.280$ and $\frac{3\sqrt{2}}{8} + \frac{1}{4} \approx 0.780$, so θ is never 0. That makes sense: $\mathbf{p}$ and $\mathbf{q}$ could only be parallel along the y-axis, where the two planes meet, but P is on that axis at $t = \frac{1}{2}\pi, \frac{3}{2}\pi$ and Q at $t = \frac{1}{4}\pi, \frac{5}{4}\pi$.
- Sampling t at two million points in $[0, 2\pi]$ and computing θ directly from the vectors gives a total time of 4.712389 with $\theta \geq \frac{1}{4}\pi$, matching $\frac{3}{2}\pi \approx 4.712389$.

Common slips

- Forgetting to divide by $|\mathbf{p}||\mathbf{q}| = 3$, or assuming $|\mathbf{q}|$ varies because the coefficients of $\mathbf{i}$ and $\mathbf{j}$ differ.
- Reversing the inequality: $\theta \geq \frac{1}{4}\pi$ means $\cos\theta \leq \frac{\sqrt{2}}{2}$, because cosine is decreasing on $[0, \pi]$.
- Counting only one period of $\cos(2t + \frac{1}{4}\pi)$: the $2t$ means it goes round twice while t goes round once.

Beyond STEP

Any circle centred at the origin can be written as $\cos s\,\mathbf{a} + \sin s\,\mathbf{b}$ with $\mathbf{a} \perp \mathbf{b}$ and $|\mathbf{a}| = |\mathbf{b}|$; if we drop those conditions we get an ellipse, and in general the image of the unit circle under a linear map. Understanding curves this way, and measuring angles between moving vectors with the scalar product, is the start of the linear algebra and vector calculus met in the first year. Strang's *Introduction to Linear Algebra* develops these ideas clearly.

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